Detection of Spatial Defect Patterns Generated in Semiconductor Fabrication Processes

Tao Yuan, *Member, IEEE,* Way Kuo, and Suk Joo Bae

***Abstract*—Spatial defect patterns generated during integrated circuit (IC) manufacturing processes contain information about potential problems in the processes. The detection of these defect patterns is crucial to improve yield and reliability in IC manufacturing. This paper proposes a multistep defect analysis approach that provides clustering results with different levels of accuracy. A defect denoising step, based on the *K*th nearest-neighbor noise removal technique, determines the existence of any clustered local defects on a wafer. If local defects exist, the denoising step separates local defects from global defects. A defect clustering step applies a similarity-based clustering technique to group the local defects into clusters according to their spatial locations. A pattern identification step identifies the pattern for each of the local defect clusters (i.e., linear, curvilinear, amorphous, or ring-shaped patterns) via various model selection criteria. Finally, a fine tuning step is applied in order to improve the accuracy of the clustering performance. The fine tuning step is based on model-based clustering with a fixed number of clusters and known patterns for each cluster. The results of both simulated and real wafer map data demonstrate the potential of our approach, both in terms of computational speed and detection accuracy, for analyzing general defect patterns generated during the IC fabrication process.**

***Index Terms*—Denoising, mixture distribution, principal curve (PC), similarity-based clustering, spatial point process, spherical shells (SS), wafer map.**

I. Introduction

THE MANUFACTURING of integrated circuits (ICs) is a complex and costly process that involves hundreds of steps and requires monitoring many key process parameters throughout the fabrication process. Defects cannot be avoided even using modern, highly automated, and precisely positioned equipment in a nearly particle-free environment [24]. A defect is defined as variation in quality that may cause a circuit failure. Defects not only lower manufacturing yield, but also cause potential reliability problems. Therefore, prompt identification of the root causes of defects and their early elimination is crucial for yield and reliability improvements in IC manufacturing [18].

The defects generated during IC fabrication are generally classified into two categories, global defects and local defects. Random causes such as particles in the cleanroom generate global defects, which are scattered all over the wafer and are very expensive to correct. Conversely, local defects are generated by assignable causes, e.g., human mistakes, particles from equipment, and chemical stains. These assignable causes are local destructive mechanisms that generate sets of aggregated defects or local defect clusters [16]. Typically, local defect clusters have amorphous, linear, curvilinear, or ring-shaped patterns [7], [24]. The specific patterns of local defect clusters reflect defect generation mechanisms. For example, particles from equipment and chemical stains may generate amorphous defect clusters [13], whereas clusters with curvilinear patterns may be caused by scratches. The spatial patterns of locally clustered defects therefore contain valuable information about defect generation mechanisms; hence, methods for detecting local defect clusters and identifying their spatial patterns are highly desirable.

A wafer map is a database system that collects and visualizes data regarding physical parameters that are measured on semiconductor wafers. Such maps contain measurement data and various metrology tools, such as ellipsometers, thickness gauges, and four point probes as well as information such as the location and size of defects. Wafer map analysis extracts and processes information about defects, such as particles, extra materials, and holes on wafers [16]. In the majority of semiconductor manufacturing processes, visual inspection of defects depends on manual review by human experts [20]. The inspectors randomly select samples from the total wafer population and try to find the root causes of defects by examining their locations, sizes, colors, and shapes using a high-resolution microscope. This human inspection procedure, however, is time consuming and highly subjective [7]. A previous study has shown agreement between observers to be as low as 45%, with long-term repeatability values of less than 95% [12]. Conversely, automated defect scanning can quickly examine a wafer using laser light. It can identify the locations of the defects and their relative sizes based on the scattering of the light. Automatic methods, capable of quickly assessing the root causes of the defects by analyzing the data obtained from automated defect scanning, are of great importance to semiconductor manufacturing because these automatic meth-

Manuscript received June 29, 2009; revised March 30, 2010; accepted April 4, 2011. Date of publication May 27, 2011; date of current version August 3, 2011. This work was supported in part by the National Science Foundation, under Projects DMI-0429176, CMMI-0825908, and CMMI-0926420. The work of S. J. Bae was supported by Mid-Career Researcher Program through NRF grant funded by the MEST, under Grant 2011-0016598.

T. Yuan is with Ohio University, Columbus, OH 43209 USA (e-mail: yuan@ohio.edu).

W. Kuo is with the City University of Hong Kong, Kowloon, Hong Kong (e-mail: way@cityu.edu.hk).

S. J. Bae is with the Department of Industrial Engineering, Hanyang University, Seoul 133-791, Korea (e-mail: sjbae@hanyang.ac.kr).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TSM.2011.2154870

0894-6507/$26.00 © 2011 IEEE

ods can lead to a considerable reduction in operator workload and improvements in accuracy and consistency. Furthermore, these methods help engineers quickly assimilate manufacturing data from defect inspection tools and then translate those data into manufacturing solutions, enhancing both yield and reliability [8], [13], [23].

There have been numerous studies on the automatic retrieval of spatial features of defect clusters in semiconductor manufacturing. For example, Gleason *et al.* [13] employed an automated clustering algorithm via artificial intelligence, Cunningham and Mackinnon [9] used an empirical clustering algorithm, while Chen and Liu [7] utilized neural networks for spatial defect pattern recognition. Many studies have assessed process variation using probe data to determine the locations of defective chips, such as [2] and [22]. However, because multiple defects on a chip result in only a single defective chip, judgments about the root causes can be made only from defective chip data. Consequently, much information is lost if the analysis is conducted using probe data for the locations of defective chips [16].

Model-based clustering has many advantages over other nonparametric clustering methods that cluster aggregated local defects, by simultaneously identifying defect clusters and obtaining spatial pattern information in the ICs yield model. In model-based clustering, observations are assumed to reflect a mixture distribution, and all of the mixing components (clusters) are usually described by identical types of models, i.e., multivariate normal distribution [3], [4]. Some studies of model-based clustering have also considered mixtures of principal curves (PCs) [21]. Hwang and Kuo [16] developed a two-step clustering strategy using model-based clustering that applies PCs to the local defects on wafer map data. However, the local defects on a wafer tend to form several patterns simultaneously. Yuan and Kuo [27], [28] considered amorphous/linear and curvilinear patterns of local defect clusters simultaneously, and modeled them using multivariate normal distributions (MVNs) and PCs, respectively, extending the traditional model-based clustering approach by considering the mixture of two different probability densities. However, their approach is mainly based on simulation results and lacks the capability to detect closed-ring shaped patterns that have been widely observed in IC manufacturing processes. Their approach is also computationally intensive when the number of defect clusters is relatively large. For instance, an IC-maker processes more than 10 000 wafers per day. One of the main bottlenecks is a probe test to check for the existence of wafer defects. The test checks for defect patterns and, if defects exist, checks the whole wafers. If even a fraction of the testing time could be reduced, it would greatly improve overall productivity.

In this paper, we propose a more flexible and less time-consuming approach for analyzing defect data. This new approach consists of multiple steps, which provide information with three different levels of accuracy. The first step, which is called "defect denoising," is based largely on the nearest-neighbor noise removal technique proposed by [5]. It provides information about whether a wafer map contains any clustered local defects. If local defects exist, it separates the global defects from the local defects. At the second step, the separated local defects are clustered together for pattern identification. This defect clustering adopts the similarity-based robust clustering technique proposed in [25]. For the third step, the spatial patterns of the defect clusters are identified using parametric models of the clustered defects. If more accurate clustering results are needed, e.g., in terms of yield estimation, fine tuning can be applied at the final step. The fine tuning step is based on model-based clustering with a new mixture distribution of three different densities: MVNs, PCs, and fuzzy spherical-shells (SSs). Unlike the mixture distribution used in [27] and [28], the number of defect clusters and the pattern of each cluster are known *a priori* in the mixture distribution to reduce the computational time required for the fine tuning step.

The rest of this paper is organized as follows. In Section II, we describe the pattern identification procedures for clustered defects: defect denoising, defect clustering, pattern identification, and fine tuning. The simulated wafer, which is shown in Fig. 1, is used in this section to illustrate our approach. More simulated wafers and real wafer map data are analyzed in Section III. A discussion and concluding remarks are presented in Section IV.

II. Sequential Pattern Identification Procedures on Clustered Defects

A. Defect Denoising

The first step in our proposed method, defect denoising, is a procedure to determine whether input defect data consist of any local defect clusters and, if local defects exist, to separate the local defects from the global defects. This step, introduced by [24], is also known as "automatic defect detection." There are a variety of different methods that can be applied, such as the spatial filter [24], the nearest-neighbor (NN) clutter removal method [5], and support vector clustering [26]. In this paper, we chose the NN clutter removal method because of its simplicity and ease of use of its tuning parameters.

We assumed that defects were distributed according to two superimposed Poisson point processes: the global defects as a Poisson point process with intensity λ_1, and the local defects as a Poisson point process with intensity $\lambda_2 (> \lambda_1)$, but restricted to some regions of the wafer and overlaid on the global defects.

The Kth NN distance D_k of a randomly chosen point defect is its distance to its kth NN defect. Thus, D_k can be modeled by mixing two transformed Gamma densities as follows:

$$D_k \sim p\Gamma^{(1/2)}(k, \lambda_1\pi) + (1-p)\Gamma^{(1/2)}(k, \lambda_2\pi) \qquad (1)$$

where p is the mixing proportion and $\Gamma^{(1/2)}$ denotes the distribution of a random variable Y where Y^2 follows a Gamma distribution. A simple application of the expectation-maximization algorithm can be used to estimate the three parameters λ_1, λ_2, and p, and to classify the defects.

Byes and Raftery [5] suggested choosing K to be the size of the smallest cluster that one wishes to detect. In our paper, we set $K = 4$ so that each local defect cluster would contain at least four defects. Fig. 2(a) and (b) shows global defects and

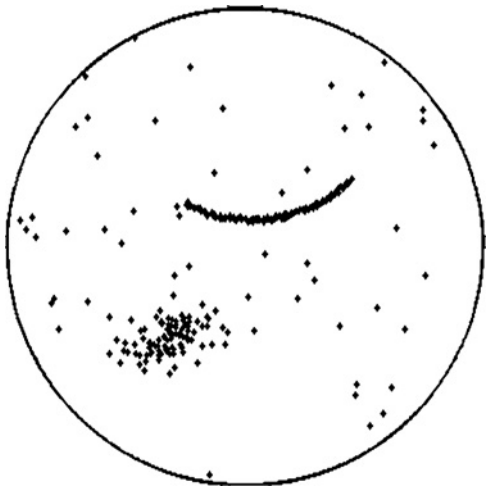

Fig. 1. Simulated defect pattern with two local defect clusters.

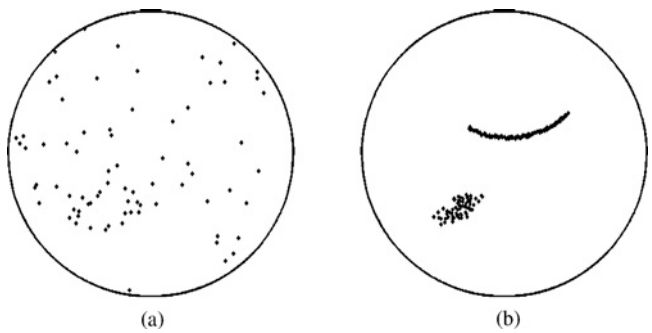

Fig. 2. Global defects and local defects identified by defect denoising for the simulated wafer shown in Fig. 1. (a) Global defects. (b) Local defects.

local defects, separated by the Kth NN denoising technique for the simulated defect data in Fig. 1.

The mixture model (1) was able to differentiate between the local and global defects using the differences in defect densities. If multiple local defect clusters have different defect densities, this denoising approach may not perform well, and we found that the spatial filter proposed by [24] may perform better in such cases if the parameters of the spatial filter are properly chosen. However, the spatial filter's performance is highly dependent on the parameters and it is usually difficult to select their values.

B. Defect Clustering

The purpose of defect clustering is to group the local defects that were detected in the defect denoising step into several clusters. In this paper, we adopted the similarity-based clustering method (SCM) proposed in [25]. The SCM method is a well-structured procedure based on a simple total similarity objective function. The SCM method is robust in terms of its initialization and outliers, and is capable of detecting clusters with different shapes. This section briefly outlines the three steps involved in the SCM.

First, the data set is denoted by $\mathbf{x} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, where $\mathbf{x}_i$ is a location vector in the d-dimensional Euclidean space $\Re^d$ and n is the total number of observations. By denoting the number of clusters as N and the center of the jth cluster as $\mathbf{c}_j$,

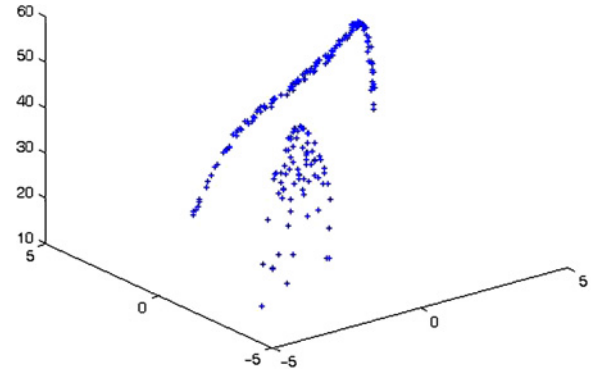

Fig. 3. 3-D plot of $\tilde{J}_s(\mathbf{x}_j)$ for the local defects shown in Fig. 2(b) with $\eta = 8$.

Yang and Wu [25] measured the similarity relation between $\mathbf{x}_i$ and $\mathbf{c}_j$ using

$$S_{ji} = \exp\left(-\frac{||\mathbf{x}_i - \mathbf{c}_j||^2}{\beta}\right) \tag{2}$$

and the total similarity using

$$J_s(\mathbf{c}) = \sum_{j=1}^{N}\sum_{i=1}^{n}\left[\exp\left(-\frac{||\mathbf{x}_i - \mathbf{c}_j||^2}{\beta}\right)\right]^{\eta} \tag{3}$$

where β is a normalized term, the power parameter η is the density shape estimation parameter, and $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_N)$. Arguing that β is not sensitive enough for classification, Yang and Wu [25] estimated β using the sample variance.

The density shape estimation parameter η determines the location and number of peaks in the objective function $J_s(\mathbf{c})$; therefore, it significantly affects the clustering results. To procure an accurate estimate for η, Yang and Wu [25] developed a correlation comparison algorithm (CCA) based on the function as follows:

$$\tilde{J}_s(\mathbf{x}_j) = \sum_{i=1}^{n}\left[\exp\left(-\frac{||\mathbf{x}_i - \mathbf{x}_j||^2}{\beta}\right)\right]^{\eta}, \quad j = 1, \ldots, n. \tag{4}$$

This function is the total similarity of the data point x_j to all of the data points. A large value for $\tilde{J}_s(x_j)$ indicates that the data point x_j is close to a cluster center and has many data points around it. The details of the CCA algorithm are summarized in Appendix A. Fig. 3 shows the 3-D plot of $\tilde{J}_s(x_j)$ for the local defects shown in Fig. 2(b). The CCA selected a value of 8 for η. There were two peaks for $\tilde{J}_s(x_j)$, indicating that there were two local defect clusters (Fig. 4).

After estimating the approximate density shape (parameter η) of the data set, the next step comprises of finding $\mathbf{c}$ in order to maximize the objective function $J_s(\mathbf{c})$ using a similarity clustering algorithm (SCA). Setting the first derivative of $J_s(\mathbf{c})$ with respect to $\mathbf{c}_j$ yields the necessary conditions for maximizing $J_s(\mathbf{c})$, namely

$$\mathbf{c}_j = \frac{\sum_{i=1}^{n} S_{ji}^{\eta}\mathbf{x}_i}{\sum_{i=1}^{n} S_{ji}^{\eta}}. \tag{5}$$

A detailed implementation of the SCA is given in Appendix B. We gave more initial cluster centers in order to process the SCA. These initial centers were then centralized to the peaks

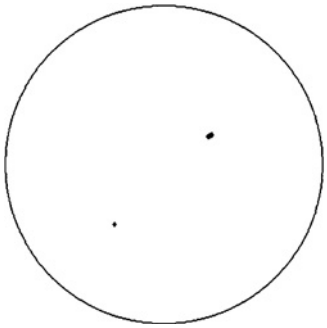

Fig. 4. Final states of the data points, initial cluster centers shown in Fig. 2(b), in defect clustering.

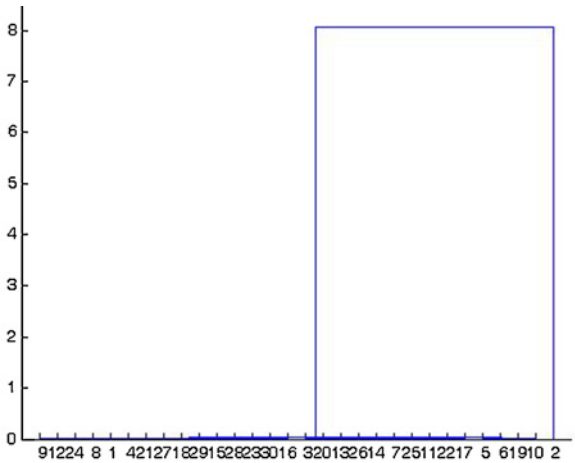

Fig. 5. Hierarchical clustering tree of the final states of the data points shown in Fig. 4.

of the SCM objective function. Fig. 4 shows the final cluster centers found by the SCA for the local defects shown in Fig. 2(b). The final cluster centers were centralized to two peaks in the SCM objective function.

If we allowed all of the data points to be the initial centers, the final SCA result had n final cluster centers (the final states of all n data points). The n final cluster centers were centralized to N peaks of the SCM objective function. To find the optimal N value and to classify the data set into these N clusters, Yang and Wu [25] applied the agglomerative hierarchical clustering (AHC) technique so as to process the n final cluster centers. There are many methods for processing AHC, such as the single linkage method, the complete linkage method, and Ward's method [25]. Fig. 5 shows the hierarchical clustering tree of the final cluster centers shown in Fig. 4. Two well-separated clusters were observed.

Fig. 6(a) shows the clustering results using the SCM in which two local defects clusters, which are distinguished using different colors and symbols in this plot, were detected.

C. Pattern Identification

The defect clustering step described in the previous section groups the local defects into several clusters according to the defect locations. To identify the pattern of each local defect cluster, we considered pattern identification to be a model-selection problem. We described different patterns using different models and determined the best model for each local defect cluster. Three different models, a multivariate normal MVN, a PC, and a SS were used to describe the distribution of local defects in a cluster with an amorphous/linear pattern, a curvilinear pattern, and a closed-ring shaped pattern, respectively.

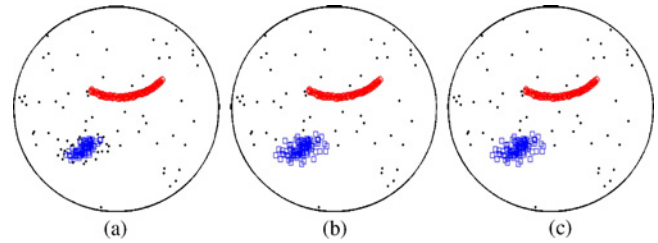

Fig. 6. Clustering results for the simulated defects shown in Fig. 1. (a) Results from the defect clustering step. (b) Results from the fine tuning step. (c) Results obtained by the model-based clustering approach of [27].

1) *Multivariate Normal Distributions:* The defects within an amorphous cluster can be easily modeled using the MVN (bivariate normal distribution for 2-D cases). The probability density function of the d-dimensional MVN is

$$f_{j,MVN}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) = (2\pi)^{-d/2}|\boldsymbol{\Sigma}_j|^{-1/2} \exp\left[-\frac{1}{2}\left(\mathbf{x}-\boldsymbol{\mu}_j\right)\boldsymbol{\Sigma}_j^{-1}\left(\mathbf{x}-\boldsymbol{\mu}_j\right)^T\right] \tag{6}$$

where $\boldsymbol{\mu}_j$ is the $(d\times 1)$ mean vector that determines the location of the jth cluster and $\boldsymbol{\Sigma}_j$ is the $(d \times d)$ variance–covariance matrix. The variance–covariance matrix is a symmetric, positive definite matrix that contains information about the shape, volume, and orientation of the cluster. A linear pattern can also be modeled by the MVN, as advocated by [3].

2) *Principal Curves:* The curvilinear feature is another typical pattern appearing across wafer defect maps. Introducing the multivariate normal distribution to describe curvilinear patterns is not ideal in that it can lead to an incorrect classification of clusters and a reduction in classification capabilities [11]. Instead, by introducing a PC, a local defect cluster with a curvilinear pattern can be efficiently modeled.

A PC is a smooth, curvilinear summary of d-dimensional data. It is a nonlinear generalization of the first principal component line as the average of the points that project to the same location on a curve. This is performed by first defining a vector function $\mathbf{f}(\gamma)$ of a scalar variable γ as a 1-D curve in d-dimensional space $\Re^d$, and $\mathbf{x}$ as a random vector in $\Re^d$. When we defined the projection index $\gamma_f(\mathbf{x}) : \Re^d \rightarrow \Re^1$, the value of γ for which $\mathbf{f}(\gamma)$ is closest to $\mathbf{x}$, as

$$\gamma_{\mathbf{f}}(\mathbf{x}) = \sup_{\gamma}\{\gamma : ||\mathbf{x} - \mathbf{f}(\gamma)|| = \inf_{\delta} ||\mathbf{x} - \mathbf{f}(\delta)||\} \tag{7}$$

thus, $\mathbf{f}(\gamma)$ was the PC (or self-consistent) of $\mathbf{x}$ if $E[\mathbf{x}|\gamma_{\mathbf{f}}(\mathbf{x}) = \gamma] = \mathbf{f}(\gamma)$ for after effects γ [14]. Here, $||\mathbf{x} - \mathbf{f}(\gamma)||$ is the Euclidean distance from the data point to its projection point on the curve. Hastie and Stuetzle [14] proposed an iterative algorithm for finding the PC $\mathbf{f}(\gamma)$.

Stanford and Raftery [21] assumed that the feature points are distributed uniformly along the true underlying feature and normally about the true underlying feature, with a mean of

zero. They argue that the probability density function of the PC is

$$f_{j,PC}(\mathbf{x}|\nu_j, \sigma_j^2) = \frac{1}{\nu_j\sqrt{2\pi}\sigma_j} \exp\left(-\frac{||\mathbf{x} - \mathbf{f}(\gamma)||^2}{2\sigma_j^2}\right) \quad (8)$$

where the scalar parameter ν_j is the length of the PC, and the scalar parameter σ_j^2 is the variance of the distances from the data points to the PC.

3) *Spherical Shells:* Defect clusters with closed-ring patterns usually reflect problems in etching or annealing processes [24]. In this paper, we introduced SSs to represent ring-shaped defect patterns. If a cluster resembles a hyper-spherical shell, then its prototype consists of two parameters $(\mathbf{c}_j, \mathbf{r}_j)$, with $\mathbf{c}_j$ being the center of the hyper-sphere and r_j its radius [17]. The hyper-spherical shell prototype at center $\mathbf{c}_j$, having radius r_j, is a set satisfying $\{\mathbf{x} \in \Re^d : (\mathbf{x} - \mathbf{c}_j)^T\mathbf{I}(\mathbf{x} - \mathbf{c_j}) = r_j^2\}$, where $\mathbf{I}$ is the identity matrix.

The squared distance from a data point at $\mathbf{x}_i$ to a prototype $(\mathbf{c}_j, \mathbf{r}_j)$ is defined as $d_{ij}^2 = \left(\left\|\mathbf{x}_i - \mathbf{c}_j\right\| - r_j\right)^2$. To find the prototype for a set of s data points, we needed to minimize the objective function $\sum_{i=1}^{s} d_{ij}^2$. Dave [10] showed that $(\mathbf{c}_j, \mathbf{r}_j)$ satisfies the two coupled nonlinear equations as follows:

$$\begin{aligned} &\textstyle\sum_{i=1}^{s} \dfrac{\left\|\mathbf{x}_i - \mathbf{c}_j\right\| - r_j}{\left\|\mathbf{x}_i - \mathbf{c}_j\right\|}(\mathbf{x}_i - \mathbf{c}_j) = 0, \\ &\textstyle\sum_{i=1}^{s} \left(\left\|\mathbf{x}_i - \mathbf{c}_j\right\| - r_j\right) = 0 \end{aligned} \quad (9)$$

which are obtained by differentiating the objective function with respect to the parameters $\mathbf{c}_j$ and r_j. Dave [10] recommended using Newton's method with initial guesses $\mathbf{c}_j = \sum_{i=1}^{s} \mathbf{x}_i$ and $r_j = \sum_{i=1}^{s} \left\|\mathbf{x}_i - \mathbf{c}_j\right\|$ to solve these two equations.

If we assume that the data points are uniformly distributed along the true underlying ring and normally about the true underlying ring, with a mean of zero, the probability density function of the spherical shell can be written as

$$f_{j,SS}(\mathbf{x}|\mathbf{c}_j, r_j, \omega_k^2) = \frac{1}{2\pi r_j\sqrt{2\pi}\omega_j} \exp\left(-\frac{(||\mathbf{x} - \mathbf{c}_j|| - r_j)^2}{2\omega_j^2}\right) \quad (10)$$

where the scalar parameter ω_j^2 is the variance of the distances from the data points to the SS.

We considered pattern identification as a model-selection problem. For each local defect cluster, we modeled the defects using the three models described above, and chose the best model according to our predefined model-selection criteria. There are many model selection rules proposed in the literature, e.g., the log_likelihood criterion [16], the Akaike information criterion (AIC) [1], and the Bayesian information criterion (BIC) [19]. The AIC and BIC are defined as follows:

$$\text{AIC}=2\log_\text{likelihood}-42m \rightarrow \text{AIC} = 2\log_\text{likelihood}-2m$$
$$\text{BIC}=2\log_\text{likelihood}-m\log(s) \rightarrow \text{BIC} = 2\log_\text{likelihood} -m\log(s)$$

where m is the number of independent parameters of a model, e.g., $m = 5$ for the bivariate normal distribution and $m = 2$ for the PC, and s is the number of defects in the cluster. The first terms of both the AIC and BIC are related to the log_likelihood, which measures the fit of a model, while the second term is the penalty for using a more complicated model. Therefore, both the AIC and the BIC balance the fit of a model and its complexity. The BIC, however, places a higher penalty on a more complicated model. Table I shows a comparison of the log_likelihood, AIC, and BIC values for the two local defect clusters shown in Fig. 6(a) when assuming different patterns. All three criteria indicate that there is an amorphous local defect cluster and a curvilinear local defect cluster on this simulated wafer.

TABLE I
IDENTIFICATION OF THE DEFECT PATTERNS SHOWN IN FIG. 7(A) USING MODEL SELECTION RULES

Cluster	Model	Model Selection Criteria		
		Log_likelihood	AIC	BIC
Curvilinear	MVN	−376.1	−762.2	−777.2
	PC	**−180.3**	**−364.6**	**−370.6**
	SS	−1079.6	−2165.2	−2174.2
Amorphous	MVN	**−127.1**	**−264.2**	**−276.0**
	PC	−165.9	−335.9	−340.6
	SS	−167.5	−341.0	−348.1

D. Fine Tuning

The defect denoising step described in Section II-A separates the global defects from the local defects based on the Kth NN distances. The defect clustering step discussed in Section II-B determines the number of local defect clusters and groups those local defects into defect clusters. Clustering errors may be induced in these two steps; therefore, in some situations, one may want to improve the accuracy of the clustering results using fine tuning techniques. One example is to estimate the yield based on the spatial distribution of the defects [15]. The fine tuning step in this section is based on a model-based clustering with a fixed number of defect clusters and known spatial patterns for each cluster.

In model-based clustering, the observations are assumed to be generated by a mixture distribution

$$f(\mathbf{x}|\boldsymbol{\Psi}) = p_0 f_0(\mathbf{x}|\boldsymbol{\theta}_0) + \sum_{j=1}^{N} p_j f_j(\mathbf{x}|\boldsymbol{\theta}_j) \quad (11)$$

where the parameters $\boldsymbol{\Psi} = (p_0, \ldots, p_N, \boldsymbol{\theta}_0, \ldots, \boldsymbol{\theta}_N)$ and the zeroth component represent the noises. The noises are usually assumed to be random and a spatial homogeneous Poisson process is used to model the distribution of the noises. The N mixing features have been predominantly modeled by multivariate normal distributions in most model-based clustering literature, i.e., $f_j(\mathbf{x}|\boldsymbol{\theta}_j)$ is the probability density function of the MVN.

Since the local defects in a curvilinear pattern may not be well modeled by the MVN, [27] modified the mixture distribution (11) by considering the mixture of two different densities, the MVN and the PC. In their mixture model, $f_j(\mathbf{x}|\boldsymbol{\theta}_j)$ is defined as

$$f_j(\mathbf{x}|\boldsymbol{\theta}_j) = f_{j,MVN}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)^{u_j} \times f_{j,PC}(\mathbf{x}|\upsilon_j, \sigma_j^2)^{1-u_j} \quad (12)$$

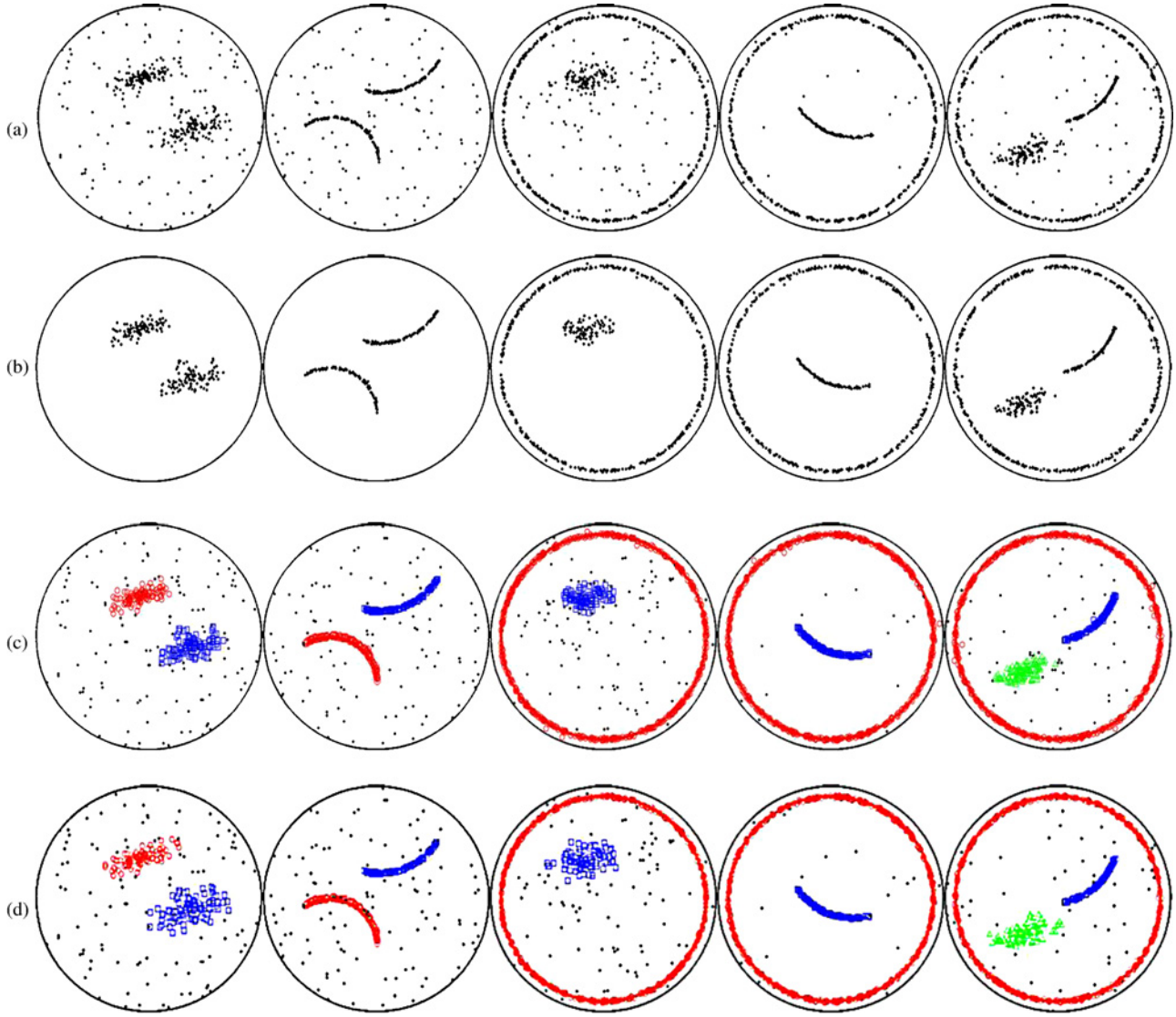

Fig. 7. Representative defect patterns generated in the simulation study. (a) Original data. (b) Local defects detected by the defect denoising step. (c) Clustering results from the defect clustering step. (d) Clustering results from the fine tuning step.

where the binary parameter u_j indicates the spatial patterns of the local defect clusters. Because the parameter u_j and the number of local defect clusters N are unknown and need to be estimated, the spatial pattern recognition approach developed by [27] may be computationally intensive when the number of defect clusters is relatively large.

In this paper, however, the number of local defect clusters N and the pattern of each cluster were determined using a defect clustering step and a pattern identification step, respectively. Therefore, the density assumed to model the defect distribution of the jth local defect cluster, $f_j(\mathbf{x}|\boldsymbol{\theta}_j)$, was known *a priori*. For example, if the jth local defect cluster has a curvilinear pattern $f_j(x|\theta_j) = f_{j,PC}(x|v_j, \sigma_j^2)$.

To estimate the model parameters, the locations $\mathbf{x}$ were regarded as having incomplete data. The complete data were considered to be $\mathbf{y}_i = (\mathbf{x}_i, \mathbf{z}_i)$, for $i = 1, \ldots, n$, where $\mathbf{z}_i = (z_{i0}, z_{i1}, \ldots, z_{iN})$ and

$$z_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_i \text{ belongs to the } j\text{th cluster,} \\ 0, & \text{otherwise.} \end{cases}$$

We assumed that $\mathbf{z}_i$ follows the identical and independent multinomial distribution of a single trial with parameters $(p_0, p_1, \ldots, p_N)$. The complete-data likelihood function is

$$L(\boldsymbol{\Psi}, \mathbf{z}|\mathbf{x}) = \prod_{i=1}^{n} \prod_{j=0}^{N} \left[p_j f_j(\mathbf{x}_i|\theta_j)\right]^{z_{ij}}. \tag{13}$$

The iterative classification-expectation-maximization (CEM) algorithm [6] was applied so as to solve the maximization problem with $L(\boldsymbol{\Psi}, \mathbf{z}|\mathbf{x})$. The lth iteration of the CEM algorithm can be summarized as follows.

1) E-step

Compute $\hat{\mathbf{z}}^{(l)}$ given $\hat{\mathbf{p}}^{(l-1)}$ and $\hat{\boldsymbol{\theta}}^{(l-1)}$. Let $t_{ij}^{(l)}$ denote the current probability of $\mathbf{x}_i$ belonging to the jth cluster as follows:

$$t_{ij}^{(l)} = \frac{\hat{p}_j^{(l-1)} f_j(\mathbf{x}_i|\hat{\theta}_j^{(l-1)})}{\sum_{m=0}^{N} \hat{p}_m^{(l-1)} f_m(\mathbf{x}_i|\hat{\boldsymbol{\theta}}_m^{(l-1)})}, \text{ for } i = 1, \ldots, n \text{ and}$$

$j = 0, 1, \ldots, N.$

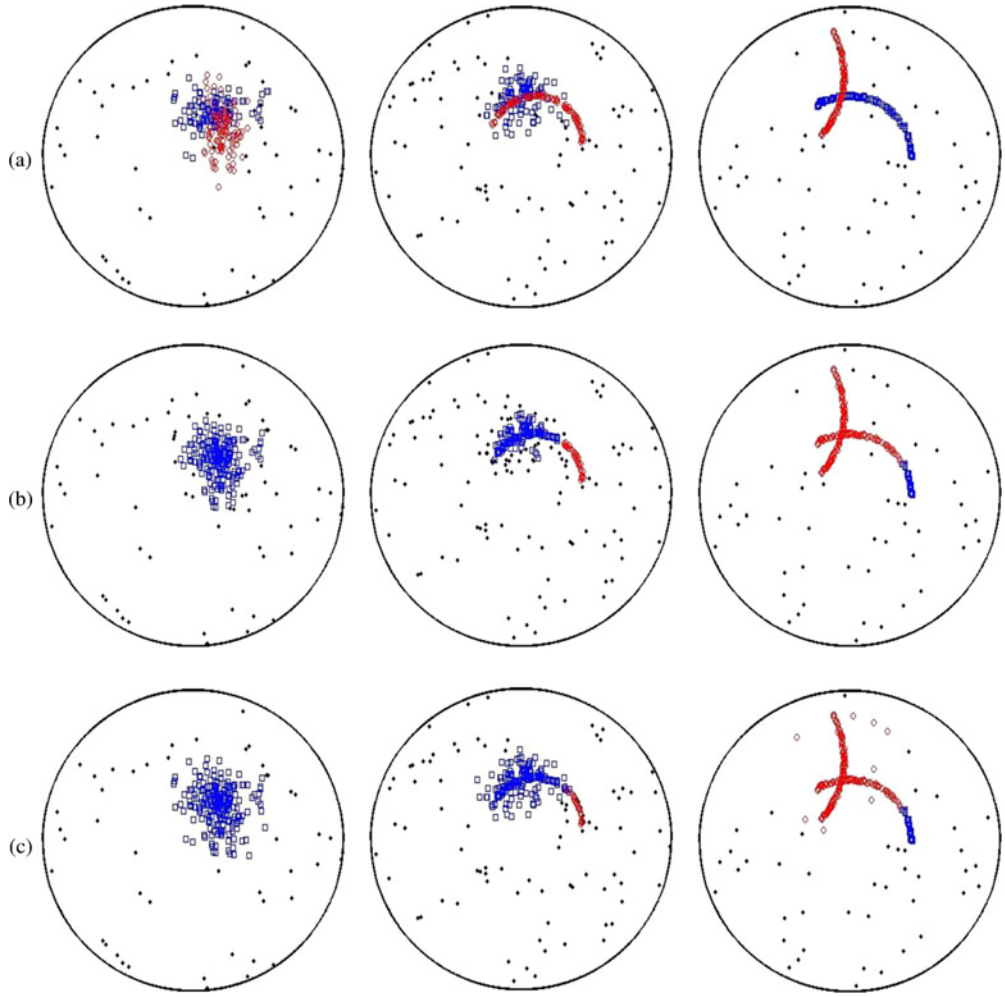

Fig. 8. Simulation study with highly overlapped clusters. (a) Original data. (b) Clustering results from the defect clustering step. (c) Clustering results from the fine tuning step.

Then $z_{ij}^{(l)} = 1$ if $t_{ij}^{(l)} = \max(t_{i0}^{(l)}, t_{i1}^{(l)}, \ldots, t_{iN}^{(l)})$ for $i = 1, \ldots, n$ and $j = 0, 1, \ldots, N$.

2) C-step
Partition $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ into $N + 1$ sets, $P_0^{(l)}, P_1^{(l)}, \ldots, P_N^{(l)}$, according to $\hat{\mathbf{z}}^{(l)}$.

3) M-step
Compute the maximum likelihood estimates $\hat{\mathbf{p}}^{(l)}$ and $\hat{\boldsymbol{\theta}}^{(l)}$. $\hat{p}_j^{(l)} = \frac{n_j^{(l)}}{n}$, where $n_j^{(l)}$ is the number of defects in the set $P_j^{(l)}$.

a) If the jth local defect cluster is modeled by a multivariate normal distribution, $\hat{\boldsymbol{\theta}}_j^{(l)} = (\hat{\mu}_j^{(l)}, \hat{\boldsymbol{\Sigma}}_j^{(l)})$, where

$$\hat{\boldsymbol{\mu}}_j^{(l)} = \bar{\mathbf{x}}_j^{(l)} = \frac{1}{n_j^{(l)}} \sum_{\mathbf{x}_i \in P_j^{(l)}} \mathbf{x}_i \quad \text{and}$$

$$\hat{\boldsymbol{\Sigma}}_j^{(l)} = \frac{1}{n_j^{(l)}} \sum_{\mathbf{x}_i \in P_j^{(l)}} \left(\mathbf{x}_i - \bar{\mathbf{x}}_j^{(l)}\right) \left(\mathbf{x}_i - \bar{\mathbf{x}}_j^{(l)}\right)^T.$$

b) If the jth local defect cluster is modeled by a PC, $\hat{\boldsymbol{\theta}}_j^{(l)} = (\hat{v}_j^{(l)}, \hat{\sigma}_j^{(l)})$, where $\hat{v}_j^{(l)}$ and $\hat{\sigma}_j^{(l)}$ can be obtained according to Section II-C.

c) If the jth local defect cluster is modeled by a SS $\hat{\boldsymbol{\theta}}_j^{(l)} = (\hat{\mathbf{c}}_j^{(l)}, \hat{r}_j^{(l)})$, where $\hat{\mathbf{c}}_j^{(l)}$ and $\hat{r}_j^{(l)}$ can be obtained according to Section II-C3.

Repeat Steps 1–3 until the convergence criterion

$$\left| \frac{\log_\text{likelihood}^{(l)} - \log_\text{likelihood}^{(l-1)}}{\log_\text{likelihood}^{(l-1)}} \right| < 0.001$$

is satisfied [6]. This convergence criterion measures the relative difference between the log_likelihood of the current iteration and that of the previous iteration. The relative tolerance 0.001 was chosen empirically. The clustering results obtained from the previous steps were used as the initial conditions for the CEM algorithm.

Fig. 6(b) shows the clustering results of the simulated wafer map data in Fig. 1 using the fine tuning step. To assess the accuracy of the clustering results, we defined the misclassification rate as follows:

$$\text{Misclassification rate} = \frac{\text{No. of wrongly clustered defects}}{\text{Total no. of defects}}.$$

The misclassification rates for the results obtained from the defect clustering step and the fine tuning step were 7.5% and

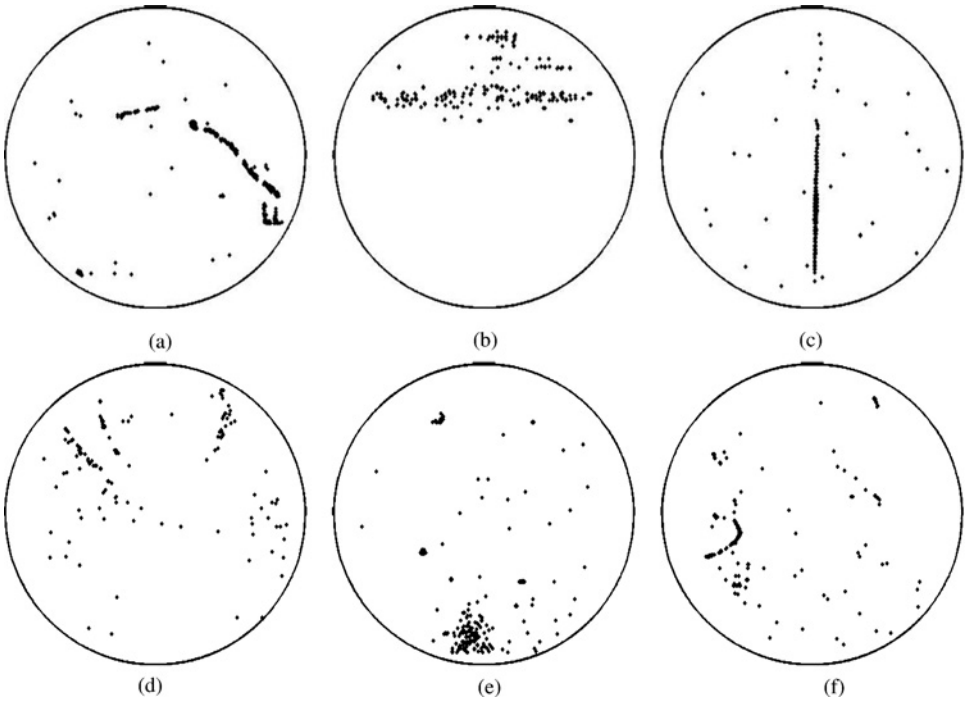

Fig. 9. Six real wafer maps collected from a DRAM manufacturer. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6.

3.6%, respectively. The improvement of accuracy can also be observed in Fig. 6(a) and (b). Due to errors in the defect denoising step, some of the local defects in the amorphous cluster were wrongly identified as global defects. The fine tuning step improved clustering results by re-clustering wrongly denoised local defects. For purposes of comparison, the clustering results from the model-based clustering approach developed by [27] were also plotted [Fig. 6(c)], and found to yield a misclassification rate of 3.1%. To compare the computational efforts required by the different methods, we used the *cputime* function of MATLAB to calculate the computational time of each individual step. The computational times for the defect denoising, defect clustering, pattern identification, and fine tuning were 3.61 s, 6.02 s, 3.67 s, and 9.67 s, respectively. The total time required by the four steps was 22.37 s. The computational time required by the model-based clustering approach by [27] was, however, 76 s. Therefore, our proposed approach greatly reduces the computational time needed to analyze defect data.

III. Results

In this section, we evaluate the performance of the proposed clustering algorithm as applied to more simulated wafers and some real wafer map data. The real wafer map data were obtained from a dynamic random access memory (DRAM) manufacturing company in Korea. For all of the cases, wafers of the same size, 20 cm in diameter, were considered. We ignored the sizes of the defects and concentrated only on locations of the defects when detecting their spatial patterns.

A. Simulation Study

We first considered more simulated wafers in order to compare the performance of the proposed method to that of other existing methods, e.g., [27]. In our study, we simulated defects on 20 wafers, including the one shown in Fig. 1. Another five representative wafers are shown in Fig. 7(a). Three types of defect patterns, i.e., amorphous, curvilinear, and closed-ring, were simulated.

The global defects were generated using a spatial homogeneous Poisson process. The numbers of global defects in each wafer were uniformly distributed between 25 and 150. The local defects in the amorphous clusters were generated by randomly sampling from the bivariate normal distributions with a random mean and a random variance-covariance matrix. The local defects in the curvilinear and ring shaped clusters were simulated by assuming that the defects were uniformly distributed along and about the curves and the circles. Each local defect cluster with an amorphous or curvilinear pattern consisted of 100 defects. Each ring-shaped cluster contained 200 defects.

Fig. 7(b)–(d) shows the results of defect denoising, defect clustering, and fine tuning, respectively. The proposed approach detected the correct number of local defect clusters and identified the correct pattern for each cluster for all of 20 simulated wafer maps. Table II summarizes the performance of the proposed approach, in addition to showing comparisons with the model-based clustering method developed by [27]. The proposed approach performed better than the model-based clustering approach of [27], in terms of both computational time and classification accuracy (15.73 versus 42.95 s for the average *cputime*, 3.6% versus 3.8% for the average misclassification rate). When calculating the misclassification rate for the model-based clustering approach of [27], we only considered wafers without any closed-ring shaped clusters, because this approach lacks the capability to detect ring-shaped clusters.

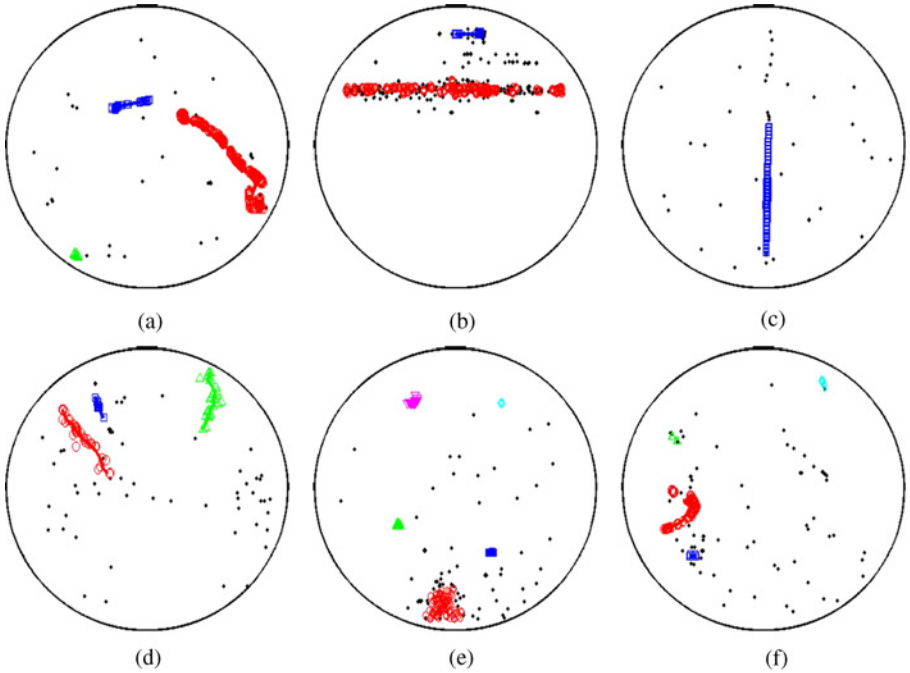

Fig. 10. Clustering results from the defect clustering step for the six wafers shown in Fig. 9. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6.

TABLE II
PERFORMANCE COMPARISONS FOR SIMULATED WAFER MAP DATA

Method		Average *cputime* (s)	Misclassification Rate		
			Average	95% Confidence Interval	Worst Case
Proposed approach	Defect denoising	4.77	–	–	–
	Defect clustering	3.57	5.9%	(1.4%, 9.4%)	9.8%
	Pattern identification	2.58	–	–	–
	Fine tuning	4.81	3.6%	(0.9%, 8.3%)	9.4%
	Total	15.73	–	–	–
Model-based clustering by [27][a]		42.95	3.8%	(2.1%, 9.0%)	9.4%

[a]Misclassification rates are calculated for simulated wafers that contain only amorphous/linear and/or curvilinear patterns.

The proposed method generally performs well when the local defect clusters are well separated. However, if two or more clusters are too close to each other or partially overlapped, the proposed method may not be able to distinguish them because this method only uses location information to analyze the clusters. Fig. 8(a) shows three simulated wafers with highly overlapped clusters. Results from the defect clustering step and the fine tuning step are shown in Fig. 8(b) and (c), respectively. The first wafer has two highly overlapped amorphous clusters. The proposed method failed to distinguish them and detected only one local defect cluster with an amorphous pattern. The second wafer has an amorphous cluster and a curvilinear cluster, which are highly overlapped. The proposed method detected two clusters and identified the correct patterns. However, the method wrongly included the left portion of the curvilinear cluster into the amorphous cluster. This resulted in very high misclassification rates of 33.45% and 28.87% for the defects clustering step and the fine tuning step, respectively. The third wafer consists of two crossed curvilinear clusters. The proposed method, although detected two local defect clusters, wrongly assigned an amorphous pattern to one of the clusters. The misclassification rates for the third wafer are 30.43% and 33.60% for the defects clustering step and the fine tuning step, respectively. The fine tuning has higher misclassification rate than the defects clustering step because of the wrongly identified defect patterns.

B. Practical Examples

Next, we analyzed some real wafer map data to determine how the proposed approach classifies general defect patterns generated during IC manufacturing. Six wafer maps with typical defect patterns were gathered from a DRAM manufacturer (Fig. 9).

Figs. 10 and 11 show the clustering results of the defect clustering step and the fine tuning step, respectively. The proposed approach demonstrated promising potential in terms of identifying the general defect patterns generated during the IC manufacturing processes.

For the purpose of comparison, Fig. 12 shows the clustering results for the six real wafer maps using the method developed by [27]. Due to their different treatments on the global defects and different criteria for determining the number of local defect clusters, the two methods may provide different clustering results for the same wafer maps. For the wafer maps

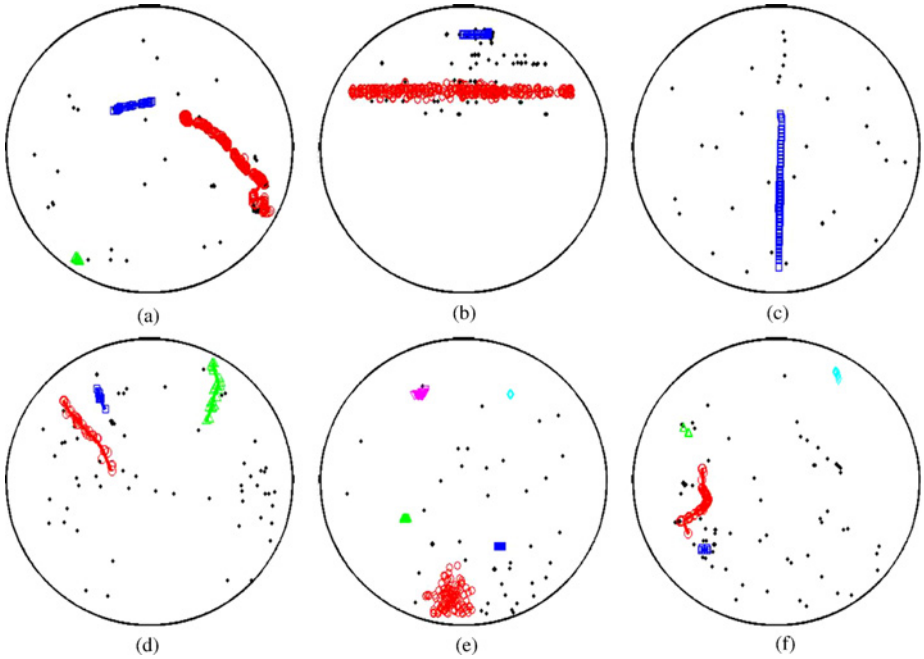

Fig. 11. Clustering results from the fine tuning step for the six wafers shown in Fig. 9. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6.

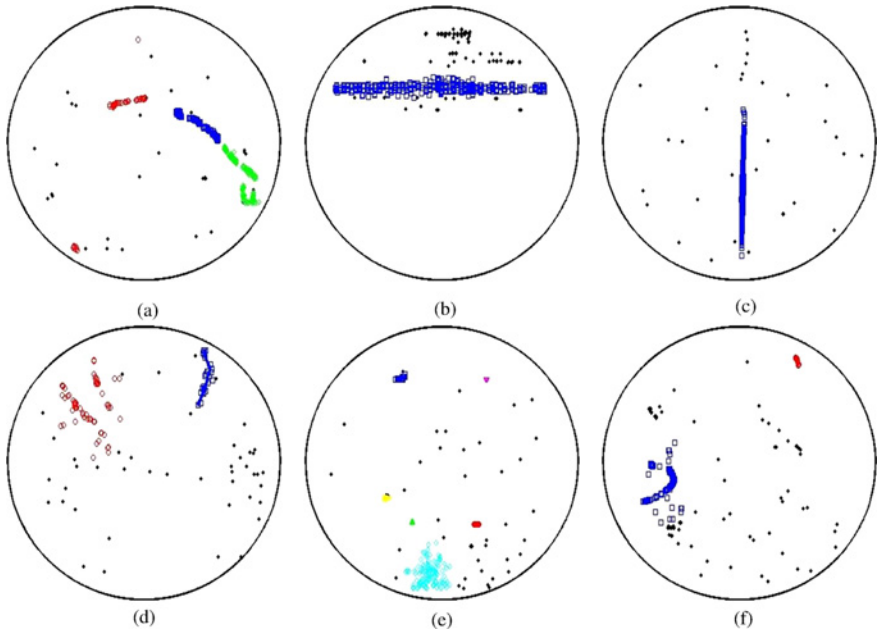

Fig. 12. Clustering results for the six wafers shown in Fig. 9 using the approach of [27]. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6.

(b), (d), and (f), the proposed sequential approach detected more clusters than Yuan and Kuo's method. For the wafer map (e), the proposed method found less clusters than the method of [27]. However, both methods are capable of capturing the significant spatial defect clusters. The newly proposed method can offer significant reduction of computational effort.

IV. Conclusion

Defect patterns on semiconductor wafers contain useful information about the causes of the defects; hence, detecting spatial defect patterns was important for integrated circuits yield and reliability improvement. This paper proposed a sequential approach for the quick detection and identification of clustered defect patterns. We customized the well-known NN clutter removal algorithm into an automatic defect scanning problem by modeling the global and local defects on the wafer via a mixture of Poisson point processes. Additionally, we modified the model-based clustering by [27] to incorporate three possible defect patterns that have been commonly observed in real IC manufacturing situations. To improve the overall performance of automatic defects classification

and pattern identification, we proposed a sequential four-step procedure. Existing model-based clustering methods [16], [27] use a single or two-step procedure for automatic defect scanning, and consider limited defects patterns customized to simulated data. We attempted to standardize the accuracy and speed of automatic defect scanning by adding defect denoising and fine tuning steps to the existing methods. By estimating the number of defect clusters and the spatial pattern of each cluster *a priori*, our optional fine tuning can refine clustering and improve clustering accuracy. Results from both simulated and real defect data demonstrate the potential of our method for detecting general defect patterns generated during the ICs fabrication processes.

The proposed method has some limitations. It uses only location information to analyze defect clusters. If two or more clusters are close to each other or partially overlap, the proposed method may not be able to accurately distinguish them. Second, the performance of the defect denoising step affects the accuracy of the clustering results. The Kth nearest-neighbor noise removal approach may not perform well in some situations, as when multiple local defect clusters on the same wafer have different defect densities. These limitations will be addressed in future research.

Acknowledgment

The author would like to thank S. J. Kim for helping them with MATLAB code.

Appendix A
Correlation Comparison Algorithm

The correlation comparison procedure for selecting a good η can be outlined as follows.

Step 1) Set $m = 1$ and specify ε_1.

Step 2) Calculate the correlation of the values of $\tilde{J}_s(\mathbf{x}_j)_{\eta_m}$ and $\tilde{J}_s(\mathbf{x}_j)_{\eta_{(m+1)}}$ for $\eta_m = m$.

Step 3) If the correlation is greater than or equal to the specified ε_1, then choose η_m to be the estimate of η, otherwise increase m by 1 and go back to Step 2.

Appendix B
Similarity Clustering Algorithm

Yang and Wu [25] developed the following iterative procedure to find c_j, which satisfies the necessary condition (5).

Initialize $\mathbf{c}_j^{(0)}$, $j = 1, \ldots, N$ and specify ε_2.

Set iteration counter $l = 0$.

Step 1) Calculate $S_{ji}^{(l+1)}$.

Step 2) Calculate $c_j^{(l+1)}$.

Repeat Steps 1 and 2 until $\max_j ||\mathbf{c}_j^{(l+1)} - \mathbf{c}_j^{(l)}|| < \varepsilon_2$.

In order to guarantee that all clusters (peaks) will be determined simultaneously, we let all data points be the initial centers, i.e., $\mathbf{c}^{(0)} = (\mathbf{c}_1^{(0)}, \ldots, \mathbf{c}_n^{(0)}) = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$ [25].

References

[1] H. Akaike, "Information theory and an extension of the maximum likelihood principle," in *Proc. 2nd Int. Symp. Inform. Theory*, 1973, pp. 267–281.

[2] S. J. Bae, J. Y. Hwang, and W. Kuo, "Yield prediction via spatial modeling of clustered defect counts across a wafer map," *IIE Trans.*, vol. 39, no. 12, pp. 1073–1083, Dec. 2007.

[3] J. D. Banfield and A. E. Raftery, "Model-based Gaussian and non-Gaussian clustering," *Biometrics*, vol. 49, pp. 803–821, Sep. 1993.

[4] H. Bensmail, G. Celeux, A. E. Raftery, and C. P. Robert, "Inference in model-based cluster analysis," *Stat. Comp.*, vol. 7, no. 1, pp. 1–10, 1997.

[5] S. Byers and A. E. Raftery, "Nearest-neighbor clutter removal for estimating features in spatial point processes," *J. Am. Stat. Assoc.*, vol. 93, no. 442, pp. 577–584, 1998.

[6] G. Celeux and G. Govaert, "A classification EM algorithm for clustering and two stochastic versions," *Comp. Stat. Data Anal.*, vol. 14, no. 3, pp. 315–332, 1992.

[7] F. L. Chen and S. F. Liu, "A neural-network approach to recognize defect spatial pattern in semiconductor fabrication," *IEEE Trans. Semiconduct. Manuf.*, vol. 13, no. 3, pp. 366–373, Aug. 2000.

[8] P. B. Chou, A. R. Rao, M. C. Sturenbecker, F. Y. Wu, and V. H. Brecher, "Automatic defect classification for semiconductor manufacturing," *Mach. Vis. Appl.*, vol. 9, no. 4, pp. 201–214, 1997.

[9] S. Cunningham and S. MacKinnon, "Statistical methods for visual defect metrology," *IEEE Trans. Semiconduct. Manuf.*, vol. 11, no. 1, pp. 48–53, Feb. 1998.

[10] R. N. Dave, "Generalized fuzzy *c*-shells clustering and detection of circular and elliptical boundaries," *Pattern Recog.*, vol. 7, no. 7, pp. 713–721, 1992.

[11] C. Fraley and A. E. Raftery, "Model-based clustering, discriminant analysis, and density estimation," Dept. Stat., Univ. Washington, Washington D.C., Tech. Rep. 329, 1998.

[12] A. Freeman, M. McIntyre, M. Retersdorf, C. Wooten, X. Song, and A. Hesse, "The application and use of an automated spatial pattern recognition (SPR) system in the identification and solving of yield issues in semiconductor manufacturing," in *Proc. IEEE/SEMI Adv. Semiconduct. Manuf. Conf.*, Jun. 2007, pp. 302–305.

[13] S. S. Gleason, K. W. Tobin, T. P. Karnowski, and F. Lakhani, "Rapid yield learning through optical defect and electrical test analysis," *Proc. SPIE-Int. Soc. Opt. Eng.*, vol. 3332, pp. 232–242, 1998.

[14] T. Hastie and W. Stuetzle, "Principal curves," *J. Am. Stat. Assoc.*, vol. 84, no. 406, pp. 502–516, 1989.

[15] J. Y. Hwang, "Yield modeling of integrated circuits based on spatial nonhomogeneous poisson process: Theory and simulation," Ph.D. dissertation, Dept. Indust. Eng., Texas A&M Univ., College Station, 2004.

[16] J. Y. Hwang and W. Kuo, "Model-based clustering for integrated circuits yield enhancement," *Eur. J. Oper. Res.*, vol. 178, no. 1, pp. 143–153, 2007.

[17] R. Krishnapuram, O. Nasraoui, and H. Frigui, "The fuzzy C spherical shells algorithm: A new approach," *IEEE Trans. Neural Netw.*, vol. 3, no. 5, pp. 663–671, Sep. 1992.

[18] W. Kuo, K. W. Chien, and T. Kim, *Reliability, Yield and Stress Burn-In: A Unified Approach for Microelectronics Systems Manufacturing and Software Development*. Boston, MA: Kluwer, 1998.

[19] G. Schwarz, "Estimating the dimension of a model," *Ann. Stat.*, vol. 6, no. 2, pp. 461–464, 1978.

[20] N. G. Shankar and Z. W. Zhong, "Defect detection on semiconductor wafer surfaces," *Microelect. Eng.*, vol. 77, no. 3, pp. 337–346, 2005.

[21] D. C. Stanford and A. E. Raftery, "Finding curvilinear features in spatial point patterns: Principal curve clustering with noise," *IEEE Trans. Pattern Mach. Intell.*, vol. 22, no. 6, pp. 601–609, Jun. 2000.

[22] W. Taam and M. Hamada, "Detecting spatial effects from factorial experiments: An approach from integrated circuit manufacturing," *Technomet.*, vol. 35, no. 2, pp. 149–160, 1993.

[23] K. W. Tobin, S. S. Gleason, T. P. Karnowski, and D. Guidry, "Using SSA to measure the efficacy of automated defect data gathering," *Micro*, vol. 16, pp. 27–33, 1998.

[24] C. H. Wang, W. Kuo, and H. Bensmial, "Detection and classification of defects patterns on semiconductor wafers," *IIE Trans.*, vol. 39, no. 12, pp. 1059–1069, 2006.

[25] M.-S. Yang and K.-L. Wu, "A similarity-based robust clustering method," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 4, pp. 434–448, Apr. 2004.

[26] T. Yuan, S. J. Bae, and J. I. Park, "Bayesian spatial defect pattern recognition in semiconductor fabrication using support vector clustering," *Int. J. Adv. Manuf. Technol.*, vol. 51, pp. 671–683, Dec. 2010.

[27] T. Yuan and W. Kuo, "A model-based clustering approach to the recognition of spatial defect patterns produced during semiconductor fabrication," *IIE Trans.*, vol. 40, no. 2, pp. 93–101, 2008.

[28] T. Yuan and W. Kuo, "Spatial defect pattern recognition in semiconductor manufacturing using model-based clustering and Bayesian inference," *Eur. J. Oper. Res.*, vol. 190, no. 1, pp. 228–240, 2008.

Tao Yuan (M'09) received the B.E. degree in thermal engineering from Tsinghua University, Beijing, China, the M.S. degree in aerospace engineering and the M.E. degree in industrial engineering from Texas A&M University, College Station, and the Ph.D. degree in industrial engineering from the University of Tennessee, Knoxville.

He is currently an Assistant Professor with the Department of Industrial and Systems Engineering, Ohio University, Columbus. His current research interests include reliability modeling and prediction, as well as reliability testing and characterization of micro/nano devices.

Dr. Yuan is a member of IIE and INFORMS.

Way Kuo is the President of the City University of Hong Kong, Kowloon, Hong Kong. He is a member of the U.S. National Academy of Engineering, Washington D.C. He has co-authored eight books on reliability and education.

He is currently the editor of IEEE TRANSACTIONS ON RELIABILITY.

Suk Joo Bae received the Ph.D. degree from the School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, in 2003.

He is currently an Assistant Professor with the Department of Industrial Engineering, Hanyang University, Seoul, Korea. From 1996 to 1999, he was a Reliability Engineer with SDI, Seoul, Korea. His current research interests include reliability evaluation of light displays, and nanodevices via accelerated life and degradation testing, statistical robust parameter design, and process control for large-volume online processing data.

Dr. Bae is a member of INFORMS, ASA, and IMS.